Lyapunov functions & Convergence of optimization algorithms (Lec8.pdf)

Ioannis Akrotirianakis

Todays topics & Logistics

- Lyapunov functions & Convergence Proofs
  - Connection between Dynamical Systems and Optimization
  - Convergence of the Steepest Descent using Lyapunov's theorem
  - Convergence rates → optional for the final exam

- **HW4 → due on Friday March 21, 23:59**
- **HW5** →out on Saturday March 22

Steepest Descent for Quadratic functions

We considered the application of Steepest Descent to Quadratic functions

$$\min_{x \in R^n} f(x) = \frac{1}{2} x^T Q x - b^T x \qquad \text{with } Q \succ 0$$

- The direction of movement at iteration k is: $d_k = -\nabla f(x_k)$ (known)

- The step size at iteration k is: $\alpha_k = \dfrac{(\nabla f(x_k))^T \nabla f(x_k)}{(\nabla f(x_k))^T Q (\nabla f(x_k))}$

- Iterative equations: $x_{k+1} = x_k - \dfrac{(\nabla f(x_k))^T \nabla f(x_k)}{(\nabla f(x_k))^T Q (\nabla f(x_k))} \nabla f(x_k)$

- **Global Convergence:** for every $x_0 \in R^n$ we have $\mathbf{log}_{k \to \infty} \boldsymbol{x_k} = \boldsymbol{x}^*$

Lyapunov Theorem – Dynamical Systems

- A **Dynamical System** (DS) is a system that changes/evolves over time
- When time takes discrete values, we have a discrete DS: $x_{k+1} = g(x_k),\ k = 0,1,2,\ldots$
- The function g is defined by the **mapping** $g: R^n \to R^n$.
- The vector $x_k \in R^n$ is called the **state** of the DS at **time** k.
- The **equilibrium** of the DS is a vector/point $x^* \in R^n$ such that: $x^* = g(x^*)$
- In **control theory** the equilibrium is taken as the origin, $x^* = 0$
- Typically, x_k is some **error** at time k and we want to drive the error to zero as time $k \to \infty$

Examples of Dynamical Systems

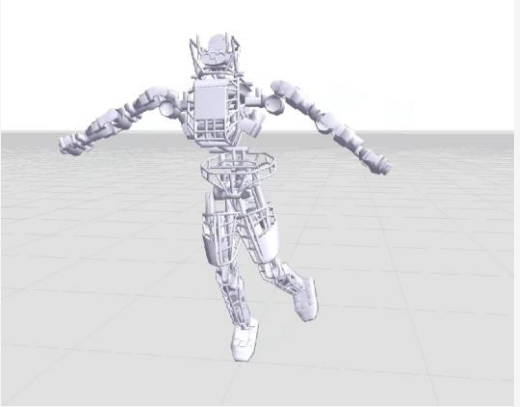
- Population of animals in a country or continent
- New cases of COVID infected people
- An autonomous car trying to stay on a lane
- An airplane trying to remain stable when it enters an area with strong wind currents
- A robot getting attacked or walks on snow and tries to keep its balance
  - Boston Dynamics products: Atlas and Handle
  - Simulations of Atlas robot trying to balance

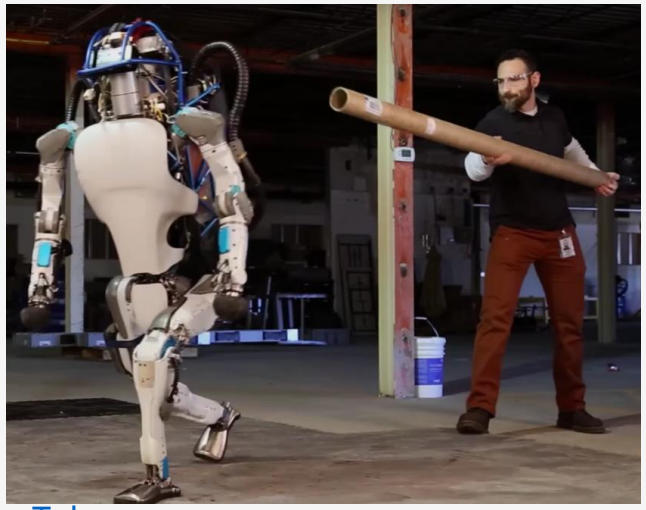
Examples of Dynamical Systems

- **Human and Robot balance** is one of the most well studied **Dynamical Systems**.

"Toyota's Latest Humanoid Robot Can Balance on One Leg", PCMag, 2017
https://www.pcmag.com/news/toyotas-latest-humanoid-robot-can-balance-on-one-leg

- Can you see where is the **equilibrium??**





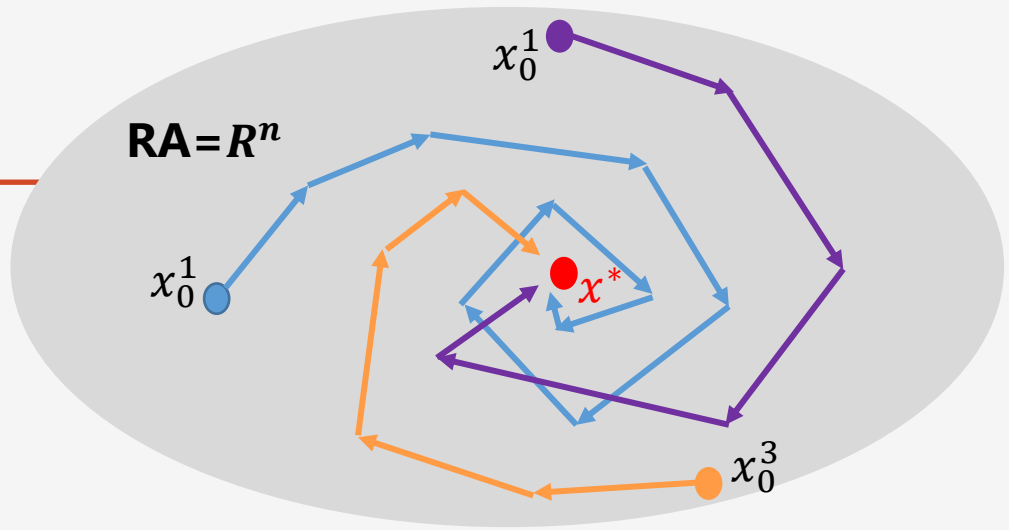
(255) Simulations of ATLAS robot balancing - YouTube

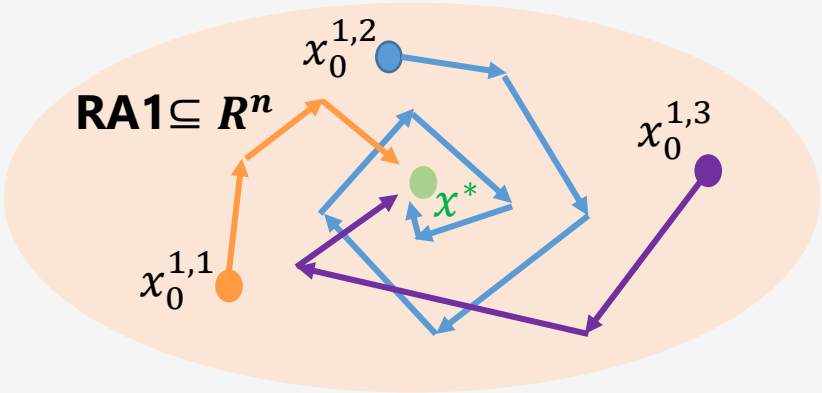
Atlas Gets a Grip | Boston Dynamics (youtube.com)

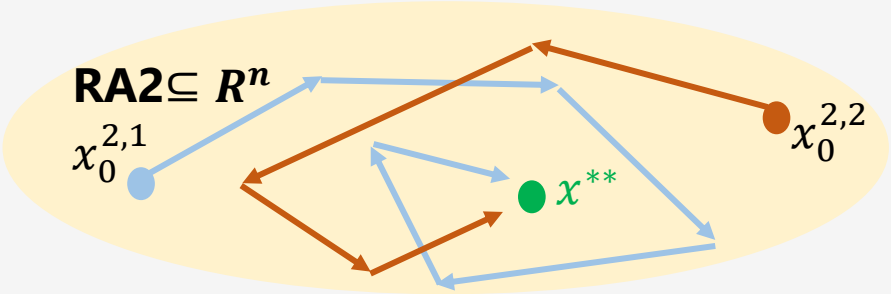
"Boston Dynamics' Atlas robot won't be pushed around",
https://www.cbsnews.com/news/boston-dynamics-new-atlas-robot/

Equilibria and Stability

- A DS may have a **unique equilibrium:**
  - In this case it is called **global equilibrium**
  - The DS is called **globally stable**
  - **Starting points**: $x_0^i \in R^n$; **<u>Region of Attraction</u> RA = R^n**
- A DS may have a **multiple equilibria:**
  - A specific equilibrium is called **local equilibrium**
  - Each equilibrium is **locally stable** with its own **RAi** $\subseteq R^n$
  - **Starting points:** $x_0^{1,i} \in$ **RA1** and $x_0^{2,i} \in$ **RA2**







Lyapunov's theorem for global stability

- **<u>Theorem:</u>** Consider the DS: $x_{k+1} = g(x_k)$, where $g: R^n \to R^n$ is a continuous map satisfying $g(0) = 0$.

- **IF** there exists a continuous function $V: R^n \to R$ satisfying the following properties:

  - <u>Prop.1:</u> $V(0) = 0$ and $V(x) > 0$, for all $x \in R^n$ and $x \neq 0$ (**positivity**)

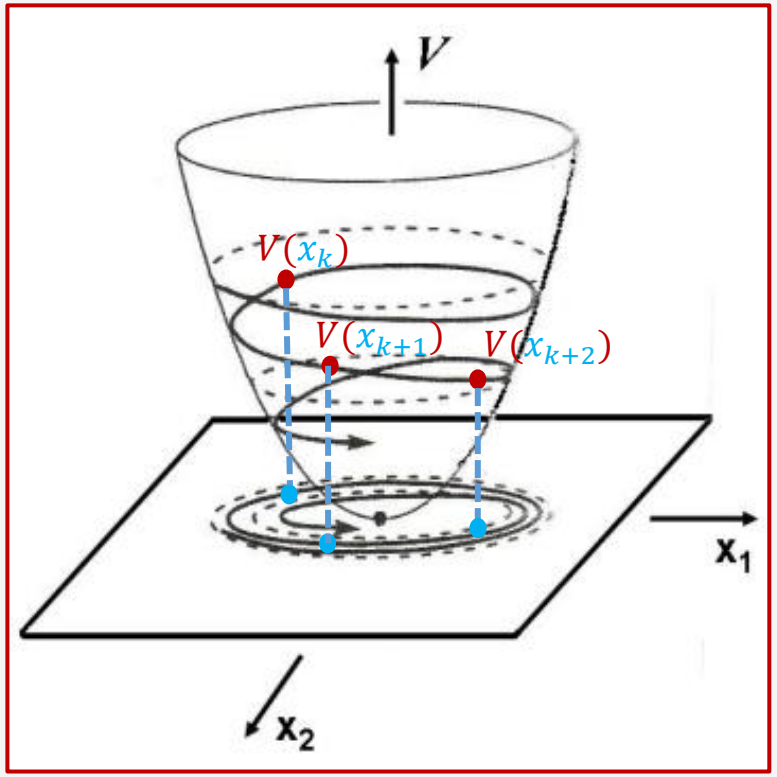
  - <u>Prop.2:</u> $V(x) \to \infty$ as $||x|| \to \infty$ (**radial unboundedness**)

  - <u>Prop.3:</u> $V(g(x)) < V(x)$, for all $x \in R^n$ and $x \neq 0$ (**decreasing)**

- **THEN** for every starting point $x_0 \in R^n$ we have $lim_{k\to\infty} x_k = 0$

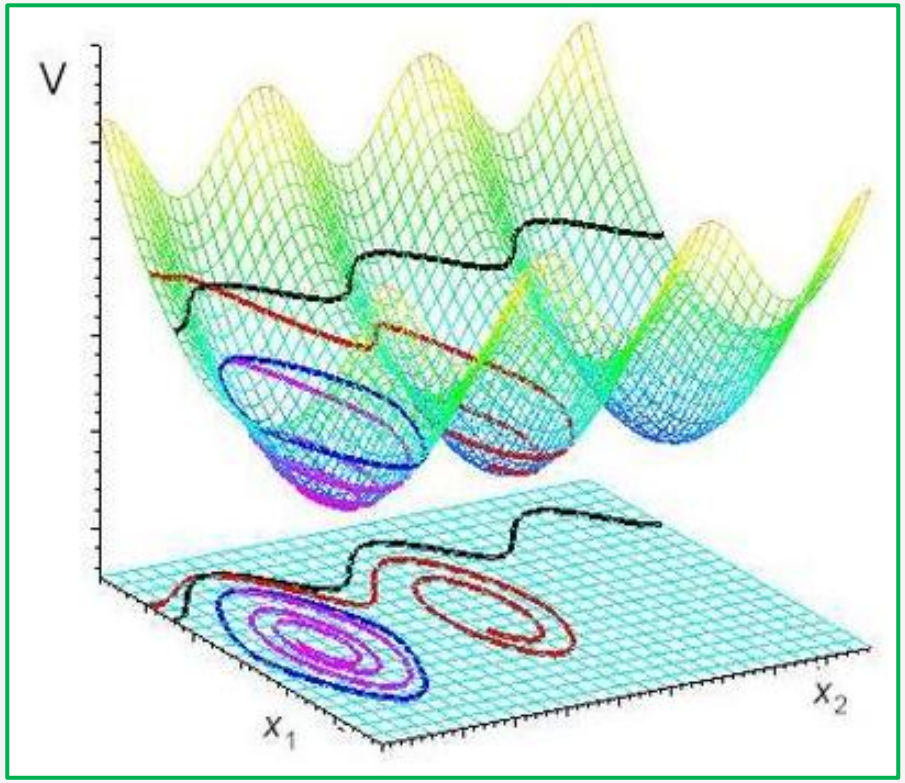
Visualizations

Lyapunov Function with a
unique equilibrium → Global stability



AAA, "*Non-monotonic Lyapunov functions for stability of nonlinear and switched systems: Theory and Computation*", MS Thesis, MIT 2008

Lyapunov Function with **many equilibria**



Firetti & Policarpi,, "*The Less Intelligent the Elements, the More Intelligent the Whole. Or, Possibly Not?*" Preprint 2020

Lyapunov's theorem for global stability

- **Proof:** According to Prop.1 the sequence $\{V(x_k)\}$ is positive and bounded from below (by 0).

- According to Prop.3 the sequence $\{V(x_k)\}$ is monotonically decreasing along all trajectories started from different x_0's.

- Hence, the sequence $\{V(x_k)\}$ converges to some constant $c \geq 0$.

  - **Case #1**: if $c = 0$ then we have $\lim_{k\to\infty} V(x_k) = 0$, and from Prop.1 we have that $x_k \to 0$ as $k \to \infty$

Lyapunov's theorem for global stability

- **Proof:** (cont.)
  - **Case #2**: if $c > 0$. We will show that this case can never happen! (we will arrive at a contradiction)
  - Indeed, if $c > 0$, then the iterates (aka, trajectory) $\{x_k : k = 0,1,2, \dots\}$ would be trapped in the compact set

$$S = \{x \in R^n \mid c \leq V(x) \leq V(x_0)\}.$$

Lyapunov's theorem for global stability

- **Proof:** (cont.)
  - Case #2: if $c > 0$. We will show that this case can never happen!
  - Indeed, if $c > 0$, then the iterates (aka, trajectory) $\{x_k : k = 0,1,2, \dots\}$ would be trapped in the compact set

    $$S = \{x \in R^n \mid c \leq V(x) \leq V(x_0)\}.$$

  - Now define: $\delta = \min_{x \in S} \left| V\big(g(x)\big) - V(x) \right|$. This is the smallest evolution between any two consecutive iterations.
    - δ exists and $\delta > 0$, because the Lyapunov function V is continuous & positive, and the set S is compact set.

Lyapunov's theorem for global stability

- **Proof:** (cont.)
  - Case #2: if $c > 0$. We will show that this case can never happen!
  - Indeed, if $c > 0$, then the iterates (aka, trajectory) $\{x_k : k = 0,1,2,\dots\}$ would be trapped in the compact set

    $$S = \{x \in R^n \mid c \leq V(x) \leq V(x_0)\}.$$

  - Now define: $\delta = \min_{x \in S} \left| V\big(g(x)\big) - V(x) \right|$. This is the smallest evolution between two consecutive iterations.
    - δ exists and $\delta > 0$, because the Lyapunov function V is continuous & positive, and the set S is compact set.
  - Hence, $V(x_k)$ decreases at every iteration by at least $d > 0$.
  - But this would imply that $\lim_{k \to \infty} V(x_k) = -\infty$, which contradicts Prop.1. Hence $c > 0$ never happens. ■

Application of Lyapunov's theorem on Steepest Descent

Connection between Dynamical Systems and Optimization

- **Theorem**: Let $\boldsymbol{f(x) = \frac{1}{2}x^T Q x - b^T x + c}$ with $\boldsymbol{Q \succ 0}$ and $x^* \in R^n$ be its **unique** minimizer. Then for every starting point $x_0 \in R^n$ the iterates $\{x_k\}$ generated by the Steepest Descent algorithm using exact line search, converge x^*.

- **Proof**: Define the **error** at time k as: $e_k = x_k - x^*$ --- $\boldsymbol{e_k}$ describes the **state** of the algorithm

- Consider the function: $V(e) = f(e + x^*) - f(x^*)$

- Let's see if it is a Lyapunov function → needs to satisfy Prop.1, Prop.2 and Prop.3

  - <u>Prop.1:</u> If $\mathrm{e} = 0$ we have: $V(0) = f(0 + x^*) - f(x^*) = 0$

    and $V(e) > 0$ for all $e \neq 0$ **WHY??**

Application of Lyapunov's theorem on Steepest Descent

- **Theorem**: Let $\boldsymbol{f(x) = \frac{1}{2}x^T Q x - b^T x + c}$ with $\boldsymbol{Q \succ 0}$ and $x^* \in R^n$ be its **unique** minimizer. Then for every starting point $x_0 \in R^n$ the iterates $\{x_k\}$ generated by the Steepest Descent algorithm using exact line search, converge x^*.

- **Proof**: Define the **error** at time k as: $e_k = x_k - x^*$ --- $\boldsymbol{e_k}$ describes the **state** of the algorithm

- Consider the function: $V(e) = f(e + x^*) - f(x^*)$

- Let's see if it is a Lyapunov function (needs to satisfy Prop.1, Prop.2 and Prop.3)

  - <u>Prop.1:</u> If $\mathrm{e} = 0$ we have: $V(0) = f(0 + x^*) - f(x^*) = 0$

    and $V(e) > 0$ for all $e \neq 0$ **WHY??** → because x^* is the unique global minimizer of f

Application of Lyapunov's theorem on Steepest Descent

- **Proof:** (cont.)

  - <u>Prop.2:</u> $V(e_k) \to \infty$ when $\left||e_k|\right| \to \infty$ **WHY??**

Application of Lyapunov's theorem on Steepest Descent

- **Proof:** (cont.)
  - <u>Prop.2:</u> $V(e_k) \to \infty$ when $||e_k|| \to \infty$ because $\boldsymbol{f}(\boldsymbol{x})$ is convex quadratic function and is unbounded from above; hence $f(x_k) - f(x^*) \to +\infty$ if $||x_k - x^*|| \to +\infty$

Application of Lyapunov's theorem on Steepest Descent

- **Proof:** (cont.)
  - <u>Prop.2:</u> $V(e_k) \to \infty$ when $||e_k|| \to \infty$ because $\boldsymbol{f}(\boldsymbol{x})$ is convex quadratic function and is unbounded from above; hence $f(x_k) - f(x^*) \to +\infty$ if $||x_k - x^*|| \to +\infty$
  - <u>Prop.3:</u> we need to show $V(e_{k+1}) < V(e_k)$

Application of Lyapunov's theorem on Steepest Descent

- **Proof:** (cont.)
  - <u>Prop.2:</u> $V(e_k) \to \infty$ when $||e_k|| \to \infty$ because $\boldsymbol{f}(\boldsymbol{x})$ is convex quadratic function and is unbounded from above hence $f(x_k) - f(x^*) \to +\infty$ if $||x_k - x^*|| \to +\infty$
  - <u>Prop.3:</u> we need to show $V(e_{k+1}) < V(e_k)$
    - $\mathrm{V}(\mathrm{e}_{\mathrm{k+1}}) =$

Application of Lyapunov's theorem on Steepest Descent

- **Proof:** (cont.)
  - <u>Prop.2:</u> $V(e_k) \to \infty$ when $||e_k|| \to \infty$ because $\boldsymbol{f}(\boldsymbol{x})$ is convex quadratic function and is unbounded from above hence $f(x_k) - f(x^*) \to +\infty$ if $||x_k - x^*|| \to +\infty$
  - <u>Prop.3:</u> we need to show $V(e_{k+1}) < V(e_k)$
    - $\mathrm{V}(\mathrm{e}_{\mathrm{k+1}}) = f(e_{k+1} - x^*) - f(x^*) =$

Application of Lyapunov's theorem on Steepest Descent

- **Proof:** (cont.)
  - <u>Prop.2:</u> $V(e_k) \to \infty$ when $||e_k|| \to \infty$ because $\boldsymbol{f}(\boldsymbol{x})$ is convex quadratic function and is unbounded from above hence $f(x_k) - f(x^*) \to +\infty$ if $||x_k - x^*|| \to +\infty$
  - <u>Prop.3:</u> we need to show $V(e_{k+1}) < V(e_k)$
    - $\mathrm{V}(\mathrm{e}_{\mathrm{k+1}}) = f(e_{k+1} + x^*) - f(x^*) = f(x_{k+1} - x^* + x^*) - f(x^*) =$

Application of Lyapunov's theorem on Steepest Descent

- **Proof:** (cont.)
  - <u>Prop.2:</u> $V(e_k) \to \infty$ when $||e_k|| \to \infty$ because $\boldsymbol{f}(\boldsymbol{x})$ is convex quadratic function and is unbounded from above hence $f(x_k) - f(x^*) \to +\infty$ if $||x_k - x^*|| \to +\infty$
  - <u>Prop.3:</u> we need to show $V(e_{k+1}) < V(e_k)$
    - $\mathrm{V}(\mathrm{e}_{\mathrm{k+1}}) = f(e_{k+1} - x^*) - f(x^*) = f(x_{k+1} - x^* + x^*) - f(x^*) =$

$$= f(x_{k+1}) - f(x^*)$$

Application of Lyapunov's theorem on Steepest Descent

- **Proof:** (cont.)
  - Prop.2: $V(e_k) \to \infty$ when $||e_k|| \to \infty$ because $\boldsymbol{f}(\boldsymbol{x})$ is convex quadratic function and is unbounded from above hence $f(x_k) - f(x^*) \to +\infty$ if $||x_k - x^*|| \to +\infty$
  - Prop.3: we need to show $V(e_{k+1}) < V(e_k)$
    - $V(e_{k+1}) = f(e_{k+1} - x^*) - f(x^*) = f(x_{k+1} - x^* + x^*) - f(x^*) =$

$$= f(x_{k+1}) - f(x^*)$$

Recall: $x_{k+1} = x_k - \alpha_k \nabla f(x_k)$ and $d_k = -\nabla f(x_k)$ is a descent direction ⇒ ??

Application of Lyapunov's theorem on Steepest Descent

- **Proof:** (cont.)
  - Prop.2: $V(e_k) \to \infty$ when $||e_k|| \to \infty$ because $\boldsymbol{f}(\boldsymbol{x})$ is convex quadratic function and is unbounded from above hence $f(x_k) - f(x^*) \to +\infty$ if $||x_k - x^*|| \to +\infty$
  - Prop.3: we need to show $V(e_{k+1}) < V(e_k)$
    - $V(e_{k+1}) = f(e_{k+1} - x^*) - f(x^*) = f(x_{k+1} - x^* + x^*) - f(x^*) =$

$$= f(x_{k+1}) - f(x^*)$$

Recall: $x_{k+1} = x_k - \alpha_k \nabla f(x_k)$ and $d_k = -\nabla f(x_k)$ is a descent direction $\Rightarrow f(x_{k+1}) < f(x_k)$

Application of Lyapunov's theorem on Steepest Descent

- **Proof:** (cont.)
  - <u>Prop.2:</u> $V(e_k) \to \infty$ when $||e_k|| \to \infty$ because $\boldsymbol{f}(\boldsymbol{x})$ is convex quadratic function and is unbounded from above hence $f(x_k) - f(x^*) \to +\infty$ if $||x_k - x^*|| \to +\infty$
  - <u>Prop.3:</u> we need to show $V(e_{k+1}) < V(e_k)$
    - $\mathrm{V}(\mathrm{e}_{\mathrm{k+1}}) = f(e_{k+1} - x^*) - f(x^*) = f(x_{k+1} - x^* + x^*) - f(x^*) =$

$$= f(x_{k+1}) - f(x^*) < f(x_k) - f(x^*)$$

Application of Lyapunov's theorem on Steepest Descent

- **Proof:** (cont.)
  - <u>Prop.2:</u> $V(e_k) \to \infty$ when $||e_k|| \to \infty$ because $\boldsymbol{f}(\boldsymbol{x})$ is convex quadratic function and is unbounded from above hence $f(x_k) - f(x^*) \to +\infty$ if $||x_k - x^*|| \to +\infty$
  - <u>Prop.3:</u> we need to show $V(e_{k+1}) < V(e_k)$
    - $\mathrm{V}(\mathrm{e}_{\mathrm{k+1}}) = f(e_{k+1} - x^*) - f(x^*) = f(x_{k+1} - x^* + x^*) - f(x^*) =$

$$= f(x_{k+1}) - f(x^*) < f(x_k) - f(x^*) = f(x_k - x^* + x^*) - f(x^*) =$$

Application of Lyapunov's theorem on Steepest Descent

- **Proof:** (cont.)
  - <u>Prop.2:</u> $V(e_k) \to \infty$ when $||e_k|| \to \infty$ because $\boldsymbol{f}(\boldsymbol{x})$ is convex quadratic function and is unbounded from above hence $f(x_k) - f(x^*) \to +\infty$ if $||x_k - x^*|| \to +\infty$
  - <u>Prop.3:</u> we need to show $V(e_{k+1}) < V(e_k)$
    - $\mathrm{V}(\mathrm{e}_{\mathrm{k+1}}) = f(e_{k+1} - x^*) - f(x^*) = f(x_{k+1} - x^* + x^*) - f(x^*) =$

$$= f(x_{k+1}) - f(x^*) < f(x_k) - f(x^*) = f(x_k - x^* + x^*) - f(x^*) =$$

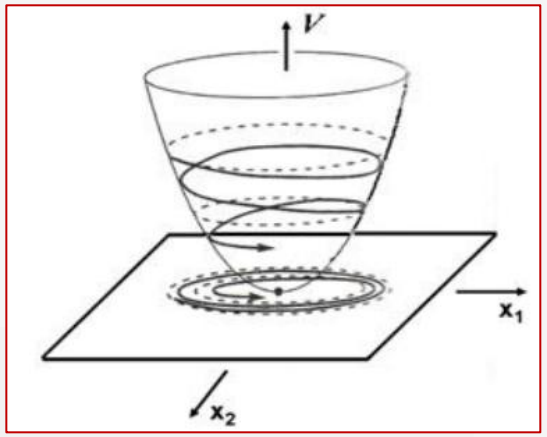
$$= f(e_k - x^*) - f(x^*)$$

Application of Lyapunov's theorem on Steepest Descent

- **Proof:** (cont.)
  - <u>Prop.2:</u> $V(e_k) \to \infty$ when $||e_k|| \to \infty$ because $\boldsymbol{f}(\boldsymbol{x})$ is convex quadratic function and is unbounded from above hence $f(x_k) - f(x^*) \to +\infty$ if $||x_k - x^*|| \to +\infty$
  - <u>Prop.3:</u> we need to show $V(e_{k+1}) < V(e_k)$
    - $V(e_{k+1}) = f(e_{k+1} - x^*) - f(x^*) = f(x_{k+1} - x^* + x^*) - f(x^*) =$

$$= f(x_{k+1}) - f(x^*) < f(x_k) - f(x^*)$$

$$= f(e_k - x^*) - f(x^*) = V(e_k)$$

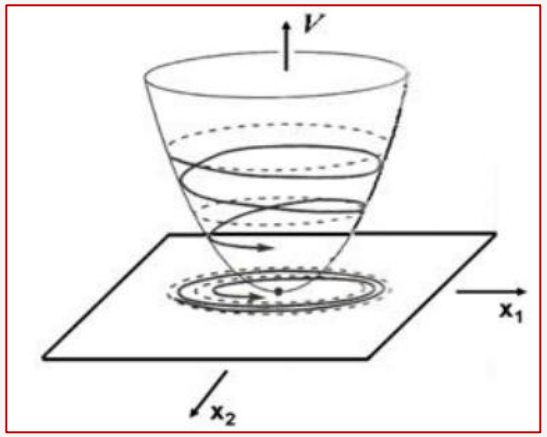


Application of Lyapunov's theorem on Steepest Descent

- **Proof:** (cont.)
  - Prop.2: $V(e_k) \to \infty$ when $||e_k|| \to \infty$ because $\boldsymbol{f(x)}$ is convex quadratic function and is unbounded from above; hence $f(x_k) - f(x^*) \to +\infty$ if $||x_k - x^*|| \to +\infty$
  - Prop.3: we need to show $V(e_{k+1}) < V(e_k)$
    - $V(e_{k+1}) = f(e_{k+1} - x^*) - f(x^*) = f(x_{k+1} - x^* + x^*) - f(x^*) =$

$$= f(x_{k+1}) - f(x^*) < f(x_k) - f(x^*) = f(x_k - x^* + x^*) - f(x^*)$$

$$= f(e_k - x^*) - f(x^*) = V(e_k)$$



- From Lyapunov's theorem: for every starting point $e_0 \in R^n$ we will have
  - $\lim_{k\to\infty} e_k = 0 \;\Rightarrow\; \lim_{k\to\infty}(x_k - x^*) = 0 \;\Rightarrow\; \lim_{k\to\infty} x_k = x^*$ ■

Application of Lyapunov's theorem for any Descent Method

Theorem: Consider the sequence of iterates $\{x_k : k = 0,1,2,\dots\}$ generated by **any** Descent Method with direction defined by $d_k = -\boldsymbol{B_k}\nabla f(x_k)$, where $\boldsymbol{B_k} \succ \boldsymbol{0}$ and the eigenvalues of B_k are larger than $\epsilon > 0$ for all iterations k. We also assume that the step size α_k is chosen by exact line search. Then the sequence $\{x_k : k = 0,1,2,\dots\}$ converges to a stationary point x^*, that is $\lim_{k\to\infty} x_k = x^*$ and $\nabla f(x^*) = 0$.

Remark: The Theorem also holds true if we use an inexact line search (e.g., Armijo line search)

Rates of convergence

Fact #1: It is very important to know that an optimization algorithm will converge to a critical point (FONC) when started from a random initial point x_0

→ It will arrive at its destination

→ Global convergence

Fact #2: It is equally important to know how fast an optimization algorithm will converge to a critical point.

→ Local convergence

→ Convergence rates allow us to estimate the **speed of the algorithm**

Rates of convergence – Orders and Factors

Assume we have a sequence $\{x_k\}$ that converges to x^*.

If $\exists k_0$ such that for all $k \geq k_0$ we have: $||x_{k+1} - x^*|| \leq \gamma ||x_k - x^*||^p$ for some $p \geq 1$ and $\gamma > 0$, then we say that the sequence $\{x_k\}$ converges to x^* with order p and factor γ.

Observations:

1. The larger the p the higher the speed

2. For fixed p, the smaller the γ the higher the speed

Rates of convergence – Orders and Factors

Some important rates of convergence

- **Linear rate**: $p = 1$ and $0 < \gamma < 1$
  - example: $\ldots \leq \left|\left|x_{k+2} - x^*\right|\right| \leq \gamma \left|\left|x_{k+1} - x^*\right|\right| \leq \gamma \left|\left|x_k - x^*\right|\right|$
- **Sublinear rate**: $p = 1$ and $\gamma = 1$ --slower than linear ($\gamma = 1$)
- **Superlinear rate**: $p > 1$ -- similar example to linear but faster
- **Quadratic rate**: $p = 2$ and $\gamma > 0$ -- usually the best rate you can hope for
  - Example: $\ldots \leq \left|\left|x_{k+2} - x^*\right|\right| \leq \gamma \left|\left|x_{k+1} - x^*\right|\right|^2 \leq \gamma \left(\gamma \left|\left|x_k - x^*\right|\right|^2\right)^2$